Micro-Expression Recognition on CASME II

Complete Results, Figures & Confusion-Matrix Walkthrough

A self-contained results document for discussion with my supervisor — every metric, every figure, and every confusion matrix explained.

Summary of the process & technologies (in brief)

We recognise **micro-expressions** (involuntary facial movements < 1/5 s) on the **CASME II** dataset. Each clip is converted into a 3-channel **motion tensor** (horizontal + vertical **optical flow** and **optical strain**), resampled to 32 frames at 224×224. A hybrid **PyTorch** network processes it: an optional **EVM** motion-magnification pre-step, a **three-stream 3-D CNN** spatial backbone, optional **SimAM** parameter-free attention, and an optional **SLSTT Transformer** temporal encoder, ending in a linear classifier. We run an **ablation study** — every on/off combination of these four components (12 valid configurations) — trained with **Focal loss** + class-balancing, and evaluated under two subject-disjoint protocols: **Leave-One-Subject-Out (LOSO)** and a **30 % hold-out**. The headline metric is **macro-F1**, because the 3 emotion classes (Negative / Positive / Surprise) are heavily imbalanced.

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1. What the experiment is

Micro-expressions are brief, involuntary facial movements that leak concealed emotion. They are faint, fast, and rare, which makes recognition hard. Instead of proposing a single model, we run an **ablation study**: four components, each switched on or off, so we can measure *which parts actually help*.

Toggle	Component	What it does
EVM	Eulerian Video Magnification	amplifies tiny, near-invisible motion before feature extraction (a data pre-step)
SimAM	parameter-free spatial attention	re-weights the important regions of the CNN feature maps (adds **0** parameters)
CNN	three-stream 3-D CNN	extracts spatial + short-range motion features — the backbone
Transformer	SLSTT temporal encoder	models long-range relationships across the 32 frames

Four on/off switches → 2^4 = 16 combinations; 4 are invalid (SimAM needs a CNN to attend to), leaving **12 configurations** we train and test.

Input = motion, not pixels. Each clip becomes a [3, 32, 224, 224] tensor of optical flow (u, v) + optical strain. Feeding motion — not RGB — isolates *how the skin deforms* (what a micro-expression actually is) and stops the model from recognising the person's face.

The data. CASME II has 255 micro-expression clips. After grouping to 3 classes and dropping the incoherent "Others" bucket, we use **156 clips from 25 subjects**:

Class	Clips	Built from raw emotions
Negative	99	disgust (63) + repression (27) + sadness (7) + fear (2)
Positive	32	happiness
Surprise	25	surprise

Why group 7 emotions into 3? Some raw classes are unusable alone — **fear has only 2 clips, sadness only 7** — you cannot train or fairly test a 2-clip class. Negative emotions (disgust/fear/sadness/repression) also share overlapping facial signatures that even human coders confuse. Grouping by valence (Negative / Positive / Surprise) is the standard MEGC challenge protocol and keeps our numbers comparable to published work.

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2. Why we report Macro-F1, not accuracy

The classes are very imbalanced (≈ 63 % Negative, 20 % Positive, 16 % Surprise). A lazy model that **always predicts "Negative"** and learns nothing scores:

"Always predict Negative"	Accuracy	Macro-F1
LOSO	**0.662**	0.266
Hold-out	**0.750**	0.286

So a useless model already gets **66–75 % accuracy**. Optimising for accuracy would *reward* ignoring the rare classes. **Macro-F1** averages each class's F1 equally, so neglecting Positive/Surprise is punished — it is the honest metric here. We therefore lead with macro-F1, keep accuracy only as a reference, and report per-class precision/recall/F1 and the full confusion matrices to show *where* errors occur.

3. The two validation methods (brief)

Both are **subject-disjoint** — the same person is *never* in both training and testing. This prevents **identity leakage** (a model recognising the *face* instead of the *expression*, which fakes the score).

- **Hold-out (fast):** split subjects **once** — ~70 % train, **30 % held out for testing** (52 test clips). One run per config; used to iterate. Downside: depends on the luck of one split (higher variance).
- **LOSO — Leave-One-Subject-Out (rigorous):** hold out **one subject**, train on the rest, test on that subject, repeat, pool predictions. It measures performance on a **completely new person** — the number that matters for real use. Downside: one full run per subject → very slow, so we ran a **pilot of 20 of the 25 subjects** (139 pooled test clips).

Why 30 % held out (and not more)? With only ~140 clips, a 10 % test set (~15 clips) is too small to measure a 3-class score and might contain zero Surprise clips. 30 % (~52 clips) is the smallest test set that keeps all three classes present and measurable. Testing on 100 % is impossible — nothing would be left to train on.

4. Results tables

4.1 LOSO (pilot, 20/25 subjects, 139 test clips) — ranked by Macro-F1

Rank	Configuration	EVM	SimAM	CNN	Transf.	Accuracy	Macro-F1
1	attention_base	–	✓	✓	–	0.643	**0.379**
2	spatial_only	–	–	✓	–	0.627	0.358
3	EVM + CNN	✓	–	✓	–	0.491	0.329
4	EVM + SimAM + CNN	✓	✓	✓	–	0.491	0.324
5	full_no_attention	✓	–	✓	✓	0.419	0.264
6	EVM + Transformer	✓	–	–	✓	0.511	0.256
7	CNN + Transformer	–	–	✓	✓	0.476	0.252
8	temporal_only	–	–	–	✓	0.600	0.251
9	full (no EVM)	–	✓	✓	✓	0.474	0.250
10	**proposed (all 4 on)**	✓	✓	✓	✓	0.413	0.249
11	pure_base	–	–	–	–	0.295	0.179
12	motion_amp_base	✓	–	–	–	0.305	0.166
—	*do-nothing baseline*					*0.662*	*0.266*

4.2 Hold-out (30 % subjects, 52 test clips) — ranked by Macro-F1

Rank	Configuration	EVM	SimAM	CNN	Transf.	Accuracy	Macro-F1
1	EVM + CNN	✓	–	✓	–	0.577	**0.458**
2	EVM + SimAM + CNN	✓	✓	✓	–	0.558	0.448
3	attention_base	–	✓	✓	–	0.481	0.402

Rank	Configuration	EVM	SimAM	CNN	Transf.	Accuracy	Macro-F1
4	spatial_only	–	–	✓	–	0.462	0.388
5	CNN + Transformer	–	–	✓	✓	0.442	0.274
6	temporal_only	–	–	–	✓	0.231	0.247
7	full (no EVM)	–	✓	✓	✓	0.308	0.218
8	EVM + Transformer	✓	–	–	✓	0.212	0.210
9	**proposed (all 4 on)**	✓	✓	✓	✓	0.173	0.183
10	full_no_attention	✓	–	✓	✓	0.192	0.176
11	pure_base	–	–	–	–	0.192	0.111
12	motion_amp_base	✓	–	–	–	0.096	0.129
—	*do-nothing baseline*					*0.750*	*0.286*

4.3 Four takeaways

1. **The 3-D CNN is essential.** Top-4 configs (both protocols) all have CNN✓ and Transformer✗. Every CNN-less config sinks to the bottom (EVM-only bottoms at 0.096 hold-out accuracy).

2. **The Transformer hurts.** In every matched pair, turning it on *lowers* macro-F1 (LOSO config_5→6: −0.13; hold-out config_13→7: −0.28). Cause: a 4-layer/8-head encoder ($\sim 10^5$ parameters) overfits on ~140 clips.

3. **More ≠ better.** The fully-loaded proposed model (all 4 on) ranks 10/12 (LOSO) and 9/12 (hold-out) — an honest negative result.

4. **EVM is protocol-dependent.** It helps hold-out (top-2 use it) but the best LOSO model omits it — magnification amplifies motion *and* noise, and the noise does not transfer to unseen subjects.

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5. The summary figures explained

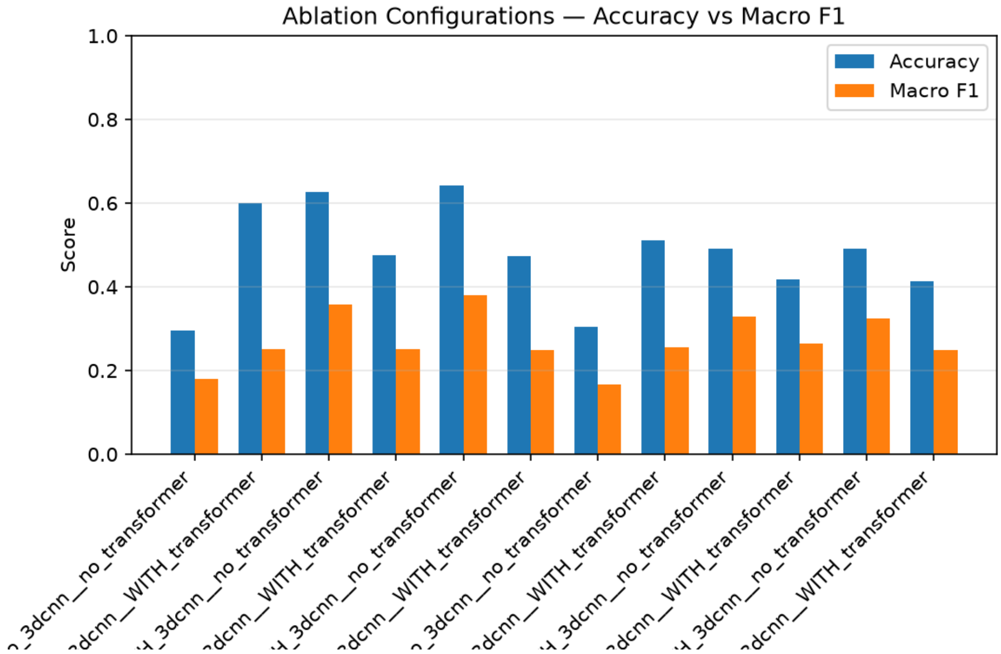
Each figure exists for both protocols (loso/plots/ and holdout/plots/).

5.1 Accuracy vs Macro-F1 (bar chart)

Axes: x = the 12 configurations; y = score 0–1. Blue = accuracy, orange = macro-F1.

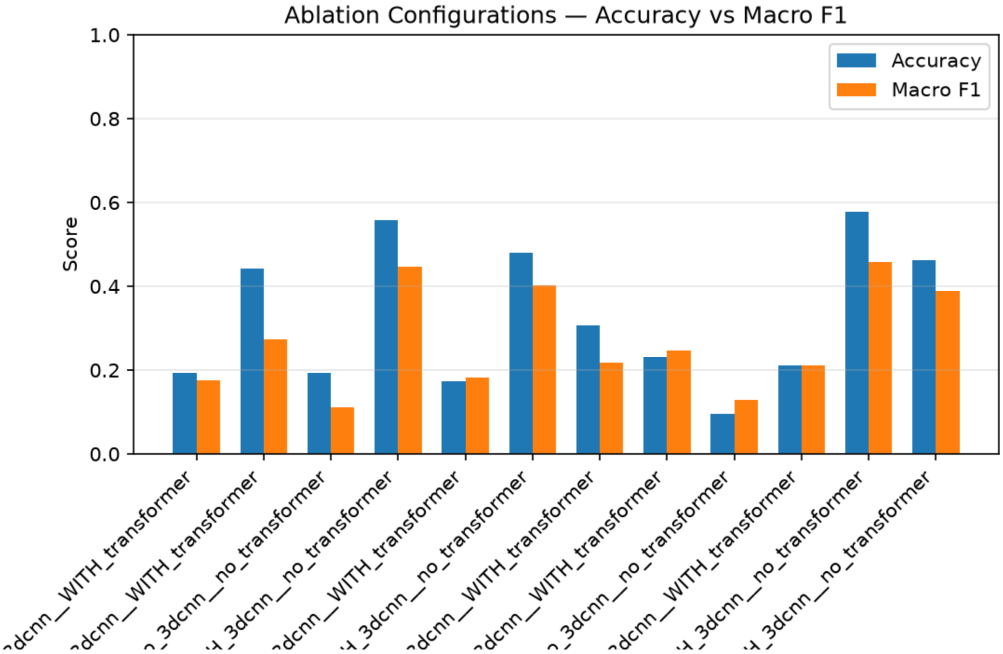
What to see: the blue (accuracy) bar is **always taller** than the orange (macro-F1) bar. That persistent gap is the fingerprint of majority-class bias — models get many Negative clips right (high accuracy) but do poorly on the rare classes (low macro-F1). This chart is the single clearest justification for leading with macro-F1.

LOSO:



LOSO — Accuracy vs Macro-F1 per configuration

Hold-out:



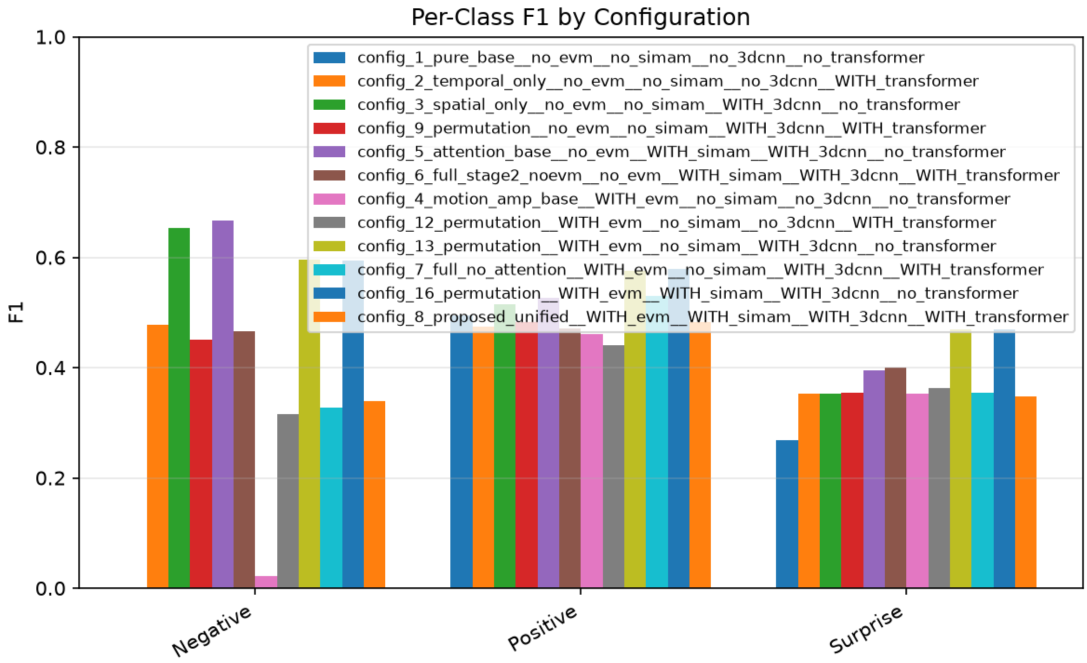
Hold-out — Accuracy vs Macro-F1 per configuration

5.2 Per-class F1 (grouped by class)

Axes: x = the three classes (Negative / Positive / Surprise); y = F1 0–1; one coloured bar per configuration.

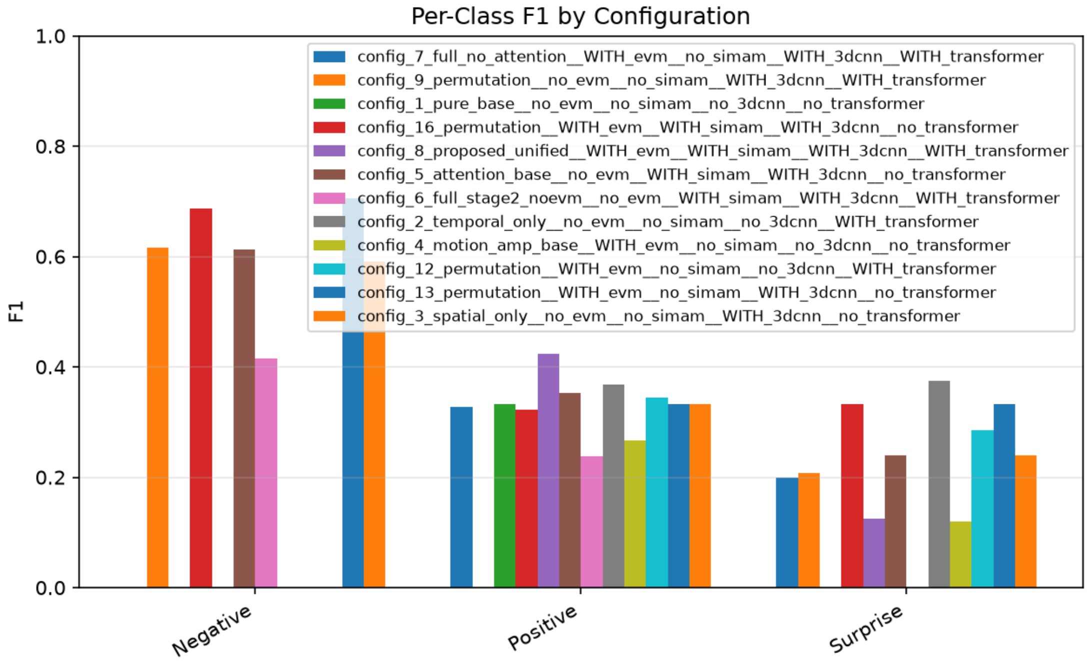
What to see: Negative F1 is high and uniform (every model finds the majority class). Positive is middling and is where good configs separate. Surprise is the lowest and most scattered — the hardest, rarest class. The ranking is decided almost entirely on Positive and Surprise, and the fact that minority bars are not flat-zero proves the imbalance defences are working. In LOSO, note config_4's near-zero Negative bar — EVM-only, CNN-less features fail even on the easy class.

LOSO:



LOSO — Per-class F1 by configuration

Hold-out:



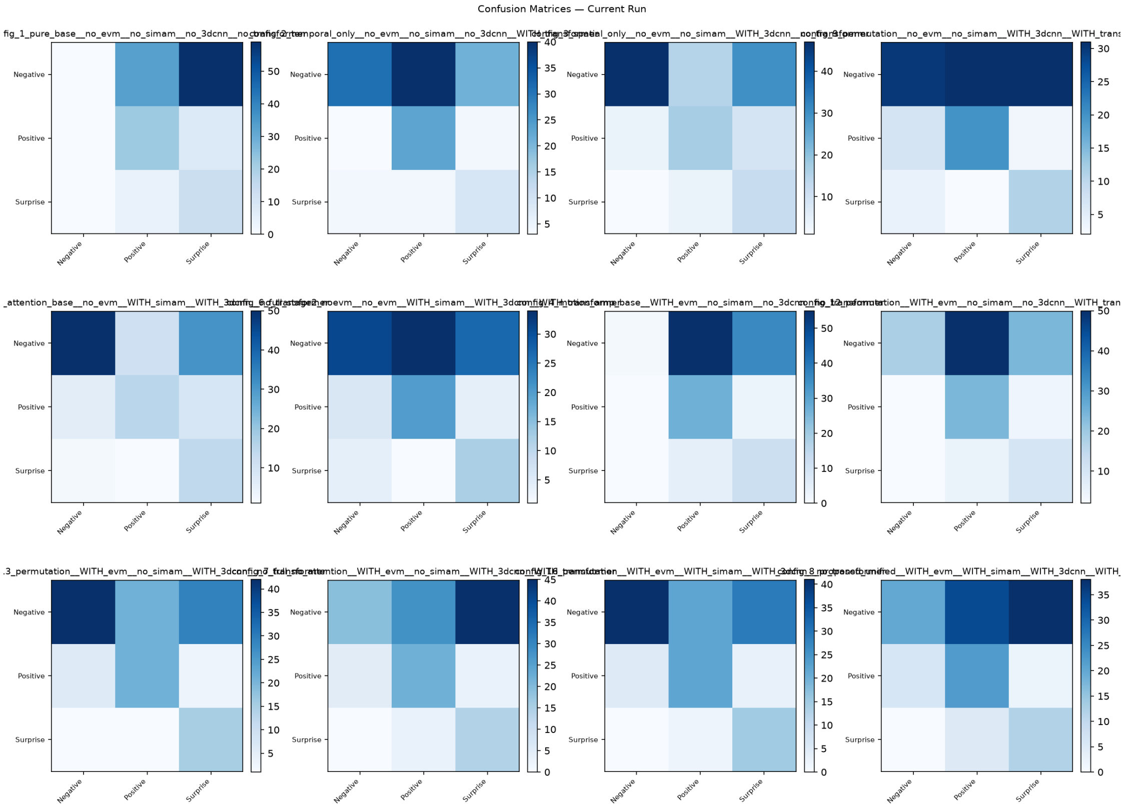
Hold-out — Per-class F1 by configuration

5.3 All confusion matrices (3×4 grid)

Axes (each small matrix): rows = true class, columns = predicted class (N / P / S); darker = more clips.

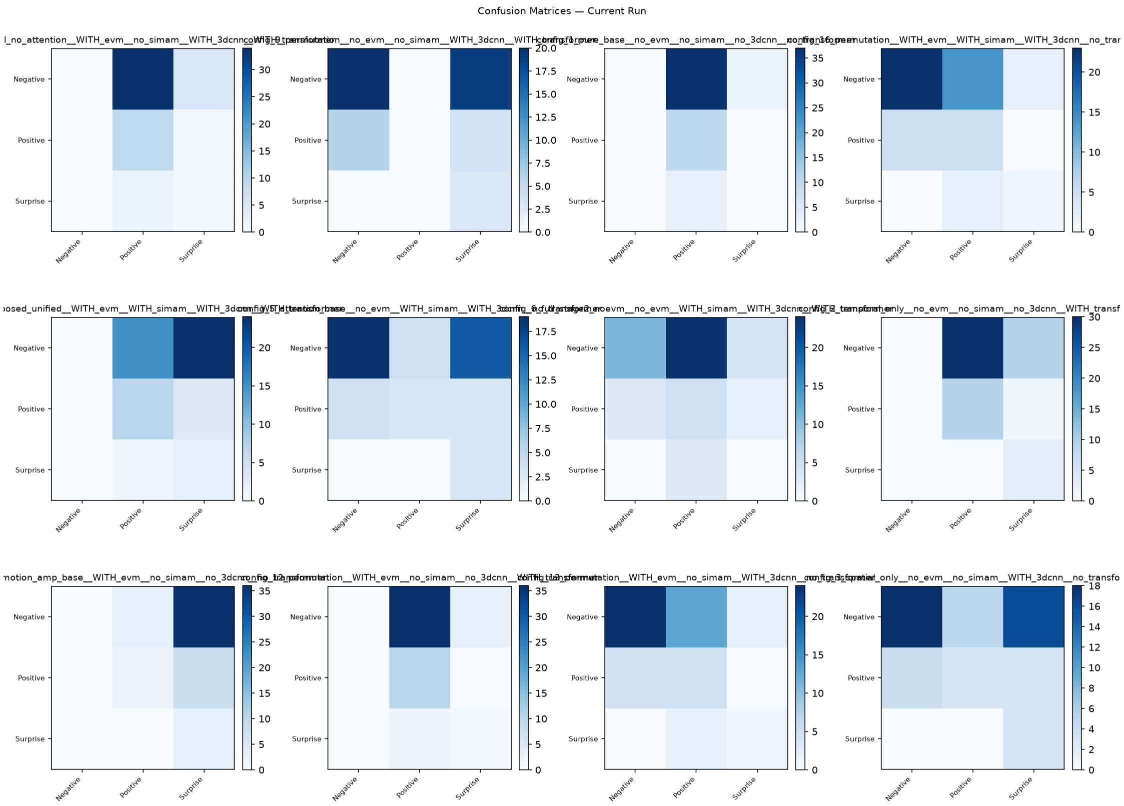
What to see: a bright **diagonal** = good; a bright **vertical stripe** = the model collapsed onto one predicted class. Good configs (3, 5, 13, 16) show a diagonal (strong top-left Negative cell). Degenerate / Transformer-heavy configs (1, 8, 4, 2) show vertical banding — e.g. config_1 predicts *no* clip as Negative (empty first column). This is the mechanism behind the F1 numbers: failures are systematic column-collapse, not random error.

LOSO:



LOSO — Confusion matrices, all 12 configs

Hold-out:

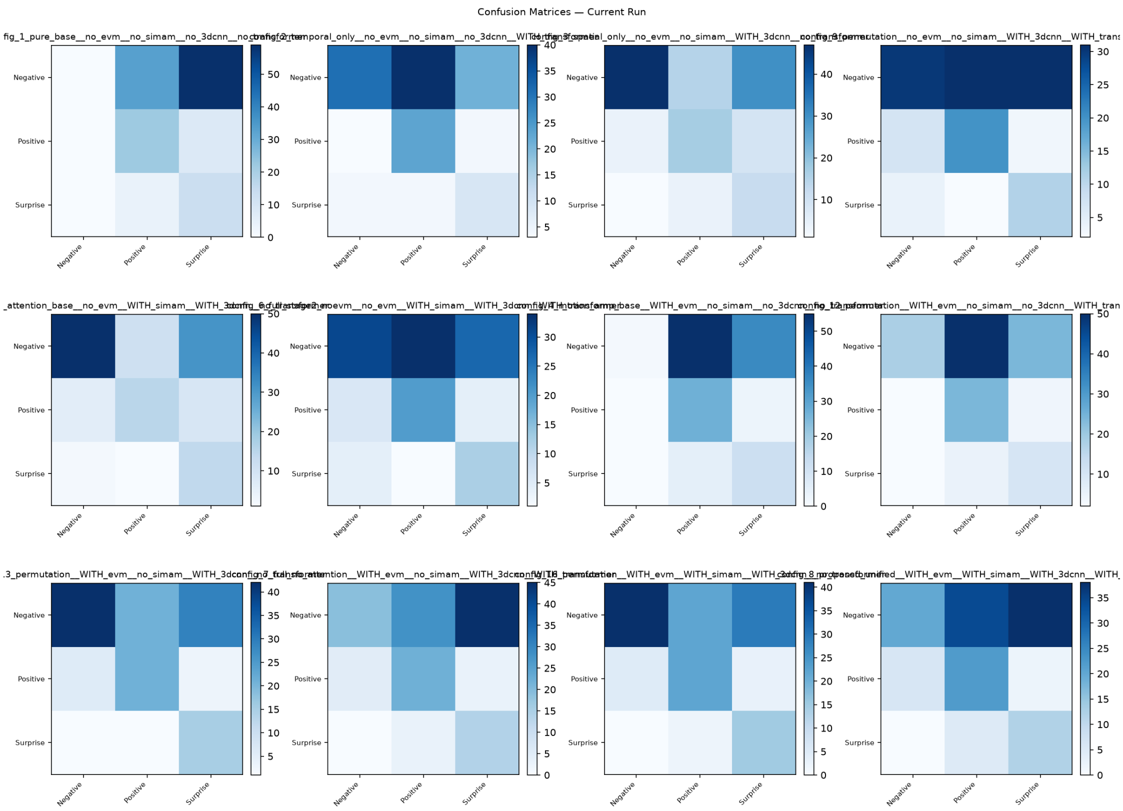


Hold-out — Confusion matrices, all 12 configs

5.4 Key configurations side-by-side

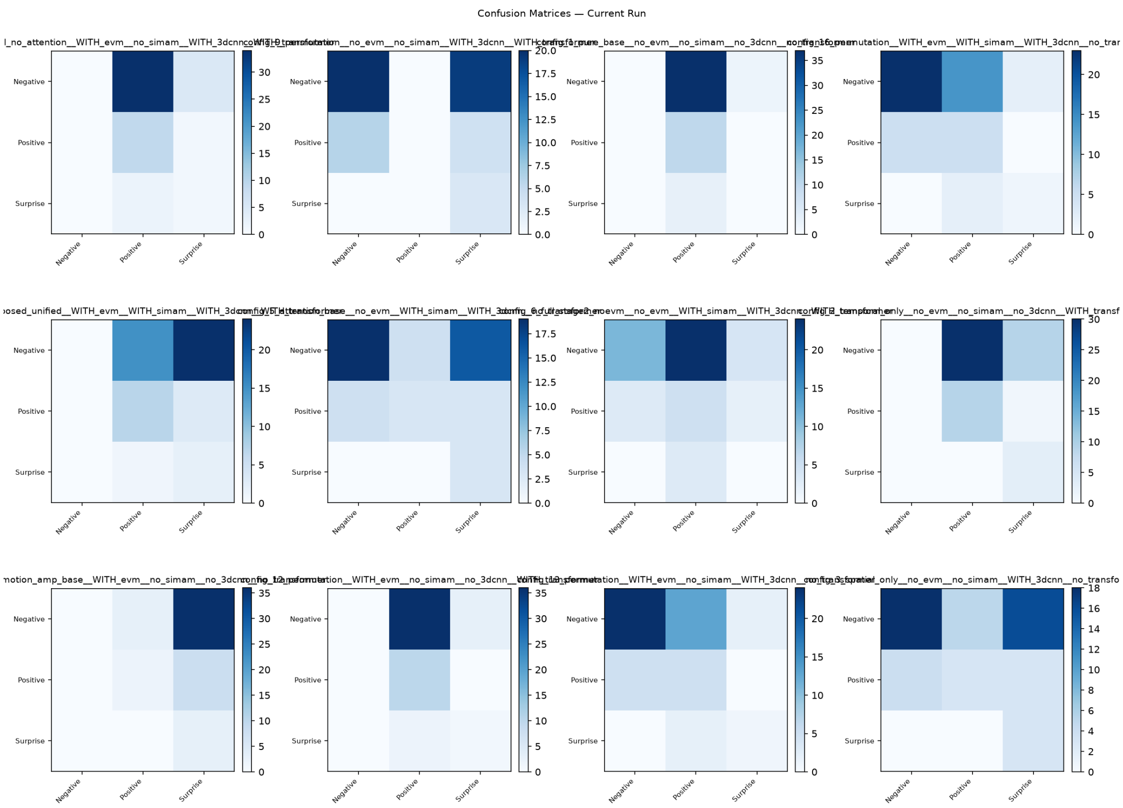
What to see: the headline configs placed adjacently. The proposed all-in model's **smeared** matrix sits next to the **clean diagonal** of the lean CNN configs — making the negative result (adding EVM + Transformer degrades a working CNN model) visually undeniable.

LOSO:



LOSO — Key configs confusion, side by side

Hold-out:



Hold-out — Key configs confusion, side by side

6. Every confusion matrix, read one by one

Below, each configuration's confusion matrix is given as three rows — the **true** class — with the three numbers being predictions for **N, P, S**. "Reading" explains the failure/success mode. (A perfect model would have all mass on the diagonal: N→N, P→P, S→S.)

6.1 LOSO — numeric confusion matrices (139 clips)

Config (E/S/C/T)	Acc	F1	True N →(N,P,S)	True P →(N,P,S)	True S →(N,P,S)	Reading
1 pure_base (....)	0.295	0.179	0,33,59	0,22,8	0,4,13	**Total majority collapse** — predicts *nothing* as Negative (empty N column). Floor baseline.
2 temporal_only (...T)	0.600	0.251	31,40,21	3,23,4	4,4,9	Over-predicts Positive; high accuracy but Negative recall only 0.34 — accuracy inflated by majority guessing.
3 spatial_only (..C.)	0.627	0.358	47,15,30	4,17,9	1,4,12	**Clear diagonal.** Strong Negative (47), leaks 30 to Surprise. Best simple CNN.
4 motion_amp (E...)	0.305	0.166	1,55,36	0,27,3	0,5,12	**Worst.** EVM without a CNN predicts almost everything Positive/Surprise; Negative recall 0.01.
5 attention_base (.SC.)	0.643	0.379	50,11,31	6,15,9	2,1,14	**Best LOSO.** Strongest Negative diagonal (50), Surprise recall 0.82. Cleanest matrix.
6 full noEVM (.SCT)	0.474	0.250	31,34,27	6,20,4	4,1,12	Adding the Transformer to config_5 **smears** the Negative row (recall 0.54→0.34).
7 full_no_attn (E.CT)	0.419	0.264	19,28,45	5,22,3	0,3,14	Negative badly leaked to Surprise (45); recall 0.21. EVM+Transformer hurts.
8 **proposed** (ESCT)	0.413	0.249	20,34,38	6,22,2	0,5,12	The full model **smears Negative** across Positive(34)/Surprise(38); recall 0.22.
9 CNN+Transf (..CT)	0.476	0.252	30,31,31	7,20,3	4,2,11	Negative split almost evenly across all three — classic Transformer smear.
12 EVM+Transf (E..T)	0.511	0.256	18,50,24	2,24,4	2,5,10	Over-predicts Positive (50); Negative recall 0.20. No CNN → weak.
13 EVM+CNN (E.C.)	0.491	0.329	42,21,29	6,21,3	1,1,15	Good diagonal; **best minority F1 of the EVM configs**; Surprise recall 0.88.
16 EVM+SimAM+CNN (ESC.)	0.490	0.324	41,22,29	5,22,3	0,2,15	Nearly identical to config_13 — SimAM adds little on top of EVM+CNN here.

6.2 Hold-out — numeric confusion matrices (52 clips)

Config (E/S/C/T)	Acc	F1	True N →(N,P,S)	True P →(N,P,S)	True S →(N,P,S)	Reading
1 pure_base (....)	0.192	0.111	0,37,2	0,10,0	0,3,0	Predicts almost everything Positive; Negative & Surprise recall 0. Degenerate.
2 temporal_only (...T)	0.231	0.247	0,30,9	0,9,1	0,0,3	No Negative predicted (empty N column) — collapse.
3 spatial_only (..C.)	0.462	0.388	18,5,16	4,3,3	0,0,3	Diagonal present; Surprise recall 1.0 but precision 0.14 (over-predicts S).
4 motion_amp (E...)	0.096	0.129	0,3,36	0,2,8	0,0,3	**Worst (acc 0.096).** Predicts nearly everything Surprise.
5 attention_base (.SC.)	0.481	0.402	19,4,16	4,3,3	0,0,3	Like config_3, slightly stronger Negative. Solid diagonal.
6 full noEVM (.SCT)	0.308	0.218	11,24,4	3,5,2	0,3,0	Transformer smears Negative into Positive(24); **Surprise F1 = 0**.

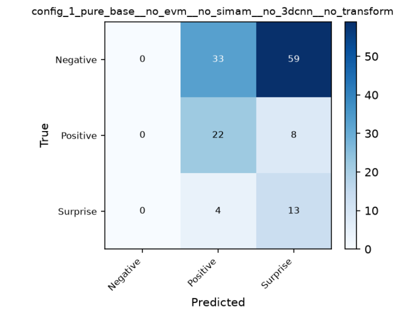
Config (E/S/C/T)	Acc	F1	True N →(N,P,S)	True P →(N,P,S)	True S →(N,P,S)	Reading
7 full_no_attn (E.CT)	0.192	0.176	0,34,5	0,9,1	0,2,1	Negative collapse (recall 0); predicts Positive.
8 **proposed** (ESCT)	0.173	0.183	0,15,24	0,7,3	0,1,2	**The full model predicts NO Negative at all** (recall 0) — worst-case majority collapse on hold-out.
9 CNN+Transf (..CT)	0.442	0.274	20,0,19	6,0,4	0,0,3	Predicts **no Positive** (empty P column); over-predicts Surprise.
12 EVM+Transf (E..T)	0.212	0.210	0,36,3	0,10,0	0,2,1	Negative collapse; predicts Positive.
13 EVM+CNN (E.C.)	0.577	0.458	24,13,2	5,5,0	0,2,1	**Best hold-out.** Strong Negative diagonal (24), balanced across classes.
16 EVM+SimAM+CNN (ESC.)	0.558	0.447	23,14,2	5,5,0	0,2,1	Near-identical to config_13 — again SimAM ≈ neutral on top of EVM+CNN.

Cross-cutting pattern: every matrix with an **empty column** is a collapsed model (it never predicts that class), and these are exactly the CNN-less or Transformer-heavy configs. Every matrix with a **populated diagonal** has the CNN on and the Transformer off. This is the same story as the ranking tables, now visible at the level of individual predictions.

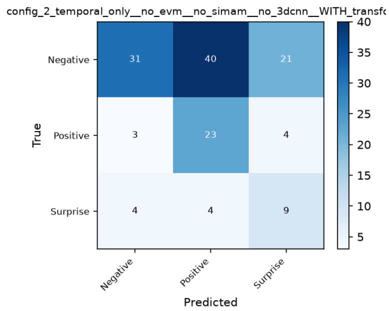
7. Appendix — individual per-config confusion matrices

Full-resolution confusion matrix for each configuration (also stored at {loso,holdout}/config_*/confusion_matrix.png).

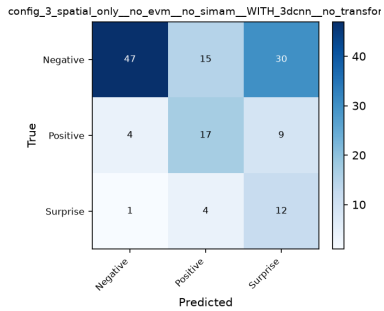
7.1 LOSO



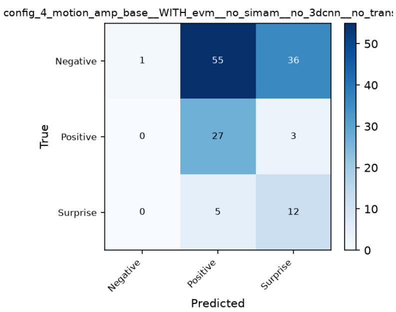
config_1 pure_base — 0.295 / 0.179



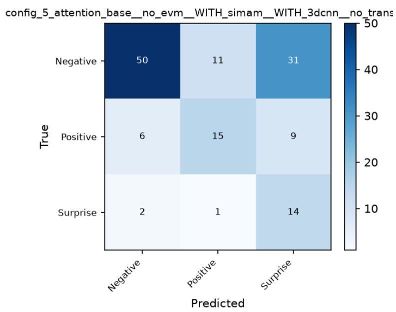
config_2 temporal_only — 0.600 / 0.251



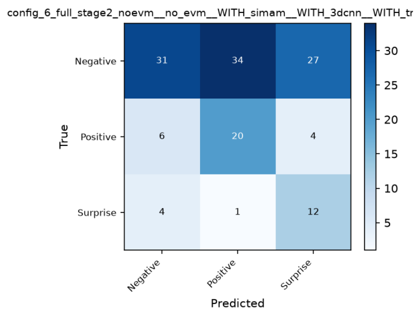
config_3 spatial_only — 0.627 / 0.358



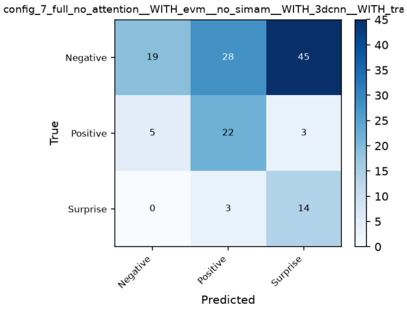
config_4 motion_amp — 0.305 / 0.166



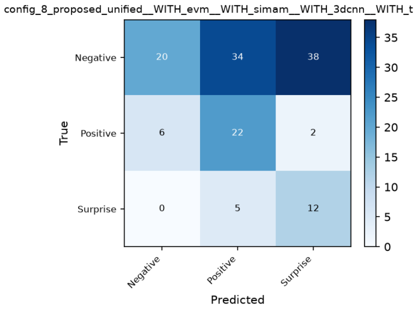
config_5 attention_base (BEST) — 0.643 / 0.379



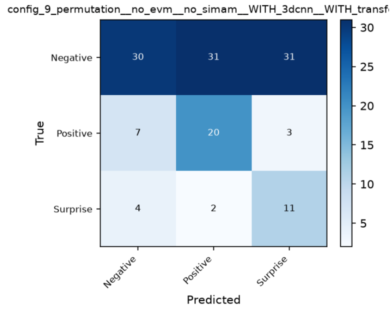
config_6 full noEVM — 0.474 / 0.250



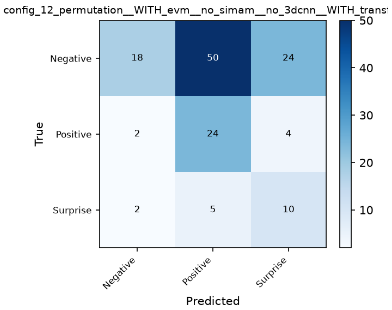
config_7 full_no_attn — 0.419 / 0.264



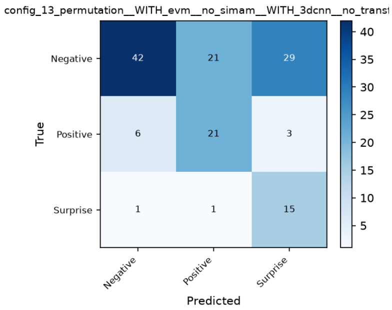
config_8 proposed (all 4) — 0.413 / 0.249



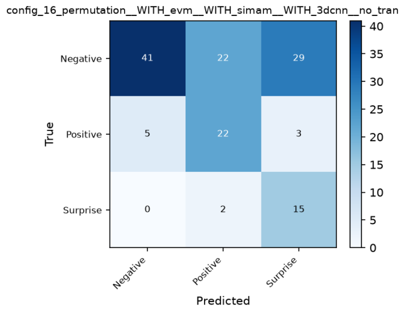
config_9 CNN+Transf — 0.476 / 0.252



config_12 EVM+Transf — 0.511 / 0.256

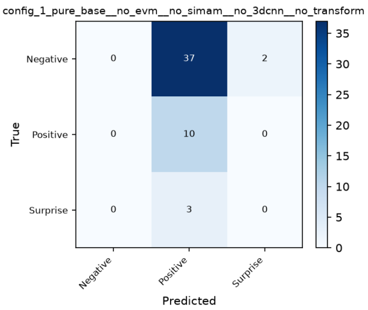


config_13 EVM+CNN — 0.491 / 0.329

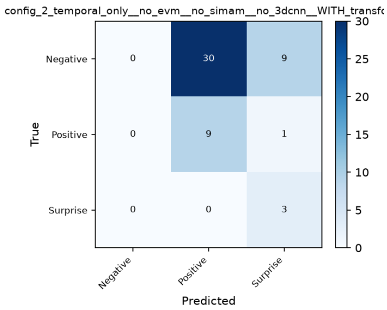


config_16 EVM+SimAM+CNN — 0.490 / 0.324

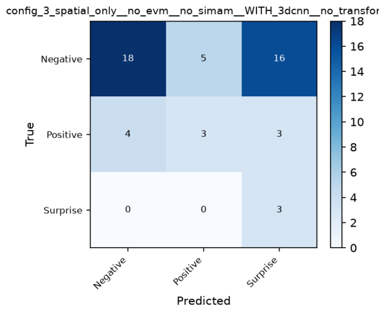
7.2 Hold-out



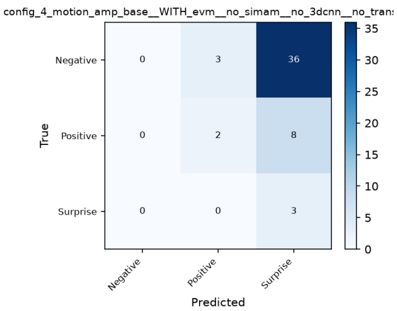
config_1 pure_base — 0.192 / 0.111



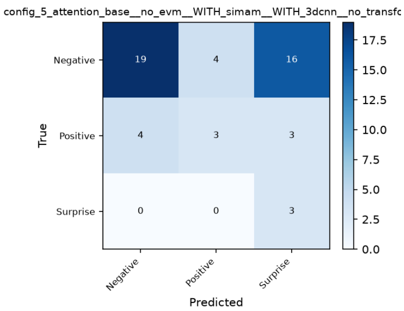
config_2 temporal_only — 0.231 / 0.247



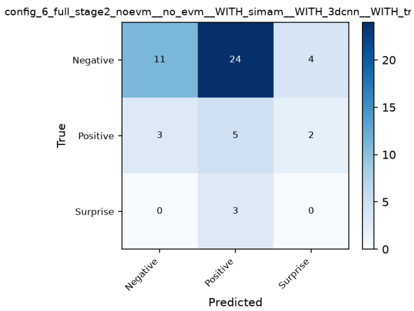
config_3 spatial_only — 0.462 / 0.388



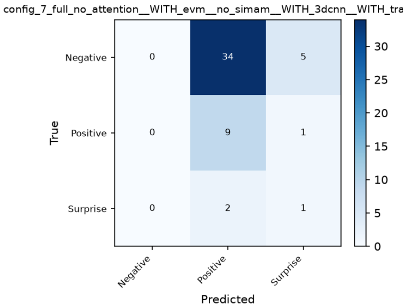
config_4 motion_amp — 0.096 / 0.129



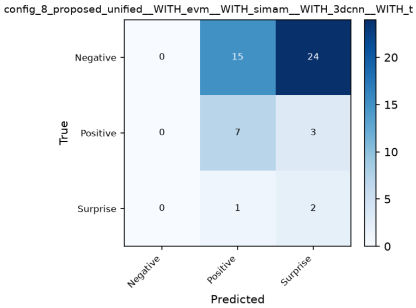
config_5 attention_base — 0.481 / 0.402



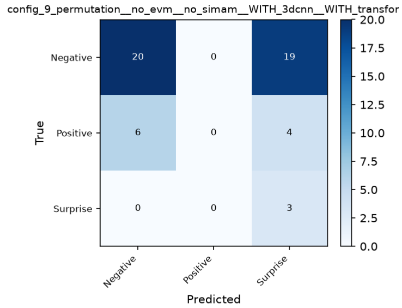
config_6 full noEVM — 0.308 / 0.218



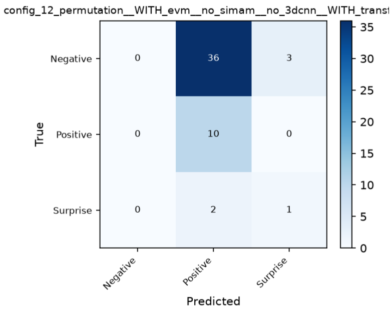
config_7 full_no_attn — 0.192 / 0.176



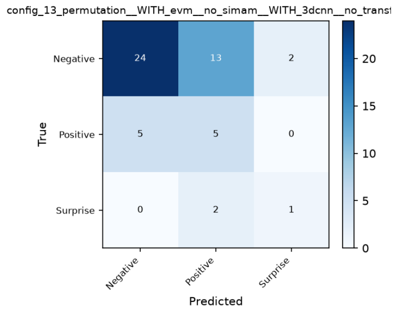
config_8 proposed (all 4) — 0.173 / 0.183



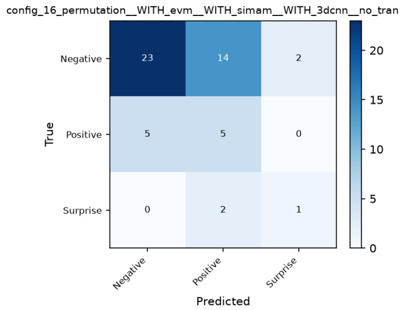
config_9 CNN+Transf — 0.442 / 0.274



config_12 EVM+Transf — 0.212 / 0.210



config_13 EVM+CNN (BEST) — 0.577 / 0.458



config_16 EVM+SimAM+CNN — 0.558 / 0.447

8. Key settings (reference)

- **Loss:** Focal loss (γ = 2) + label smoothing 0.05 + inverse-frequency class weights + balanced oversampling (three stacked defences against the 63/20/16 imbalance).
- **Optimiser:** AdamW, lr 1e-4, weight decay 1e-4, gradient clipping 1.0, mixed precision, CosineAnnealing schedule.
- **Batch size 2** — a hardware limit (a [3,32,224,224] clip needs ~5 GB VRAM for Transformer configs).
- **Best checkpoint selected by macro-F1** (not accuracy); seed 42 for reproducibility.

All numbers from results_weekend/{loso,holdout}/summary.csv and each config's final_results.json; class counts from master_thesis_labels.csv; 30 % hold-out from gui_settings.json.